

Translated

From: e-maxx.ru

Hungarian algorithm for solving the assignment problem

Statement of the assignment problem

There are several standard formulations of the assignment problem (all of which are essentially equivalent). Here are some of them:

- There are n jobs and n workers. Each worker specifies the amount of money they expect for a particular job. Each worker can be assigned to only one job. The objective is to assign jobs to workers in a way that minimizes the total cost.
- Given an $n \times n$ matrix A , the task is to select one number from each row such that exactly one number is chosen from each column, and the sum of the selected numbers is minimized.
- Given an $n \times n$ matrix A , the task is to find a permutation p of length n such that the value $\sum A[i][p[i]]$ is minimized.
- Consider a complete bipartite graph with n vertices per part, where each edge is assigned a weight. The objective is to find a perfect matching with the minimum total weight.

It is important to note that all the above scenarios are "**square**" problems, meaning both dimensions are always equal to n . In practice, similar "**rectangular**" formulations are often encountered, where n is not equal to m , and the task is to select $\min(n, m)$ elements. However, it can be observed that a "rectangular" problem can always be transformed into a "square" problem by adding rows or columns with zero or infinite values, respectively.

We also note that by analogy with the search for a **minimum** solution, one can also pose the problem of finding a **maximum** solution. However, these two problems are equivalent to each other: it is enough to multiply all the weights by -1 .